

OPTIMIZING CREDIT ALLOCATION AND STRESS- TESTING BANK EXPOSURE UNDER SEMICON 2.0:

**A Machine Learning Framework for
Macro-Prudential Risk Management**

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Problem & Research Gap

India's semiconductor ecosystem is expanding rapidly, with projects involving very large and diverse capital requirements. Banks may face concentrated exposure to the sector, yet the sector has very limited historical credit outcome data, making conventional supervised default prediction methodologically unreliable.

Concentrated Bank Exposure

Lenders risk significant sectoral concentration without adequate tools to differentiate structural risk profiles.

Data Scarcity

Historical credit outcomes are nearly absent, rendering standard supervised ML classification unsuitable.

The Core Problem: Credit Outcome Audit:

01

36 Semiconductor Projects

Full universe of identified projects in India

02

Credit Outcome Audit

Systematic search for verifiable default or distress records

03

Only 2 / 36 Verifiable Outcomes

Both showed no financial distress

04

Conventional Default ML Unsuitable

Insufficient outcome labels for supervised classification

05

Alternative Risk Framework Needed

Unsupervised clustering + stress-testing approach adopted

Research Objectives & Progress

- 1 Data Collection & Cleaning**
36-project database assembled and standardised
- 2 Target Feasibility Audit**
Confirmed supervised ML is not viable; alternative approach justified
- 3 Supervised ML Benchmark**
Established baseline to demonstrate limitations of conventional methods
- 4 PCA + K-Means Clustering**
Dimensionality reduction and structural segmentation completed
- 5 Cluster Validation**
Cross-method and bootstrap validation performed
- 6 Stress Testing**
Baseline, mild, moderate, and severe macro scenarios applied
- 7 Monte Carlo Simulation**
10,000 iterations to assess tail risk and ranking stability
- 8 Credit Allocation**
Risk-adjusted allocation with diversification constraints modelled
- 9 Banking Decision Support**
Risk grades, early warning flags, and sensitivity analysis delivered

Building the Semiconductor Risk Dataset

Dataset Composition:

36 PROJECTS

Total 12 Manufacturing + 24 Design / DLI projects form the complete universe of India's identified semiconductor initiatives.

Feature Categories:

Capital

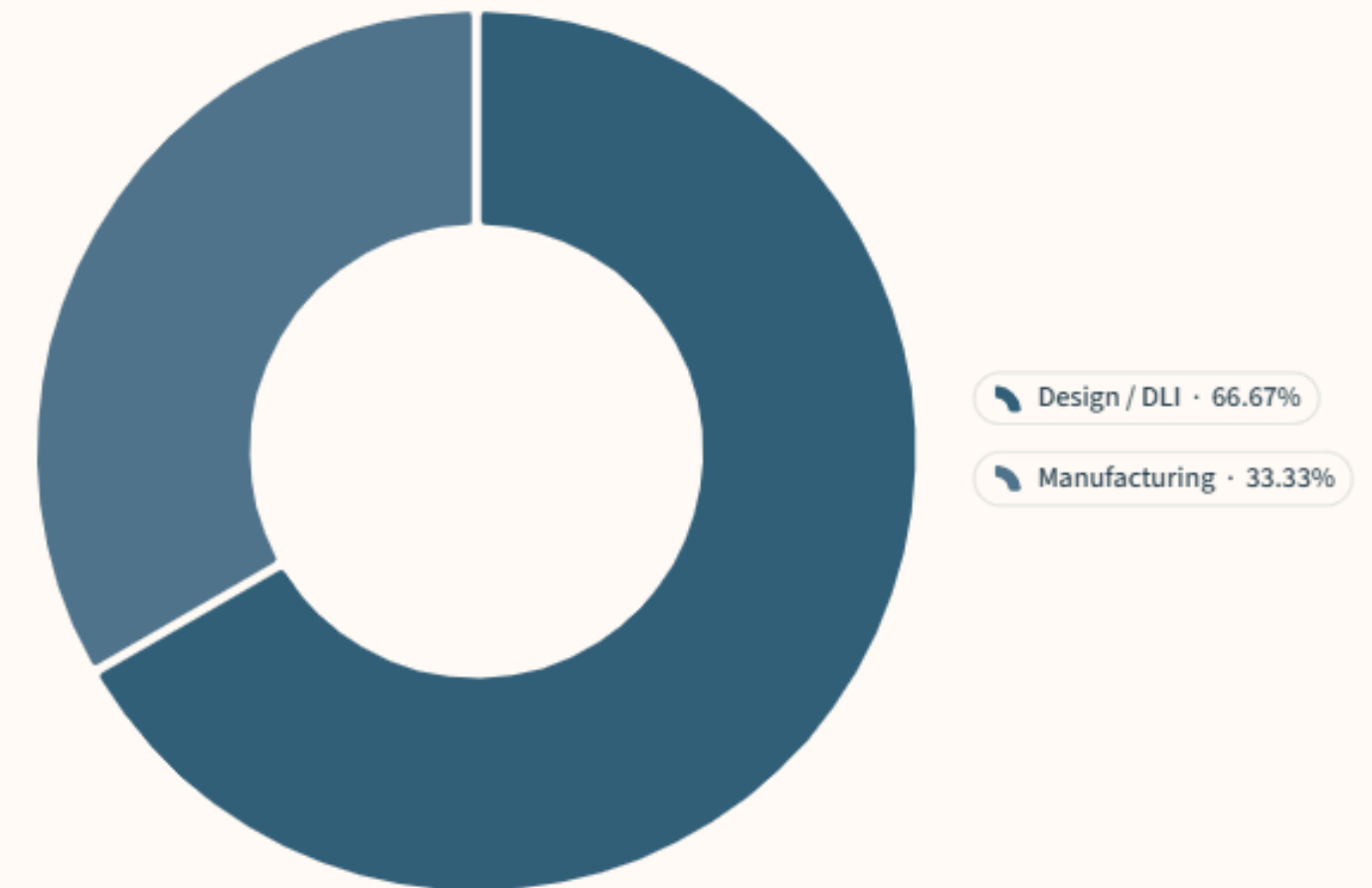
Total investment / CapEx;
log-scaled investment to
handle scale disparity

Technology

Wafer fabrication, OSAT /
packaging, Compound
semiconductor, Fabless
design

Macro-Financial

RBI credit growth, gross NPA, manufacturing / IIP indicators



Design/DLI projects account for two-thirds of the dataset; manufacturing projects form the basis for stress testing.

From High-Dimensional Data to Structural Segments ML

ML Methodology

12 Features

Capital, technology, location, and macro-financial inputs

7 Principal Components

Retained after PCA dimensionality reduction

85.06% Variance

Proportion of total information preserved in the reduced space

K = 6 Clusters

Optimal number of structural segments identified via K-Means

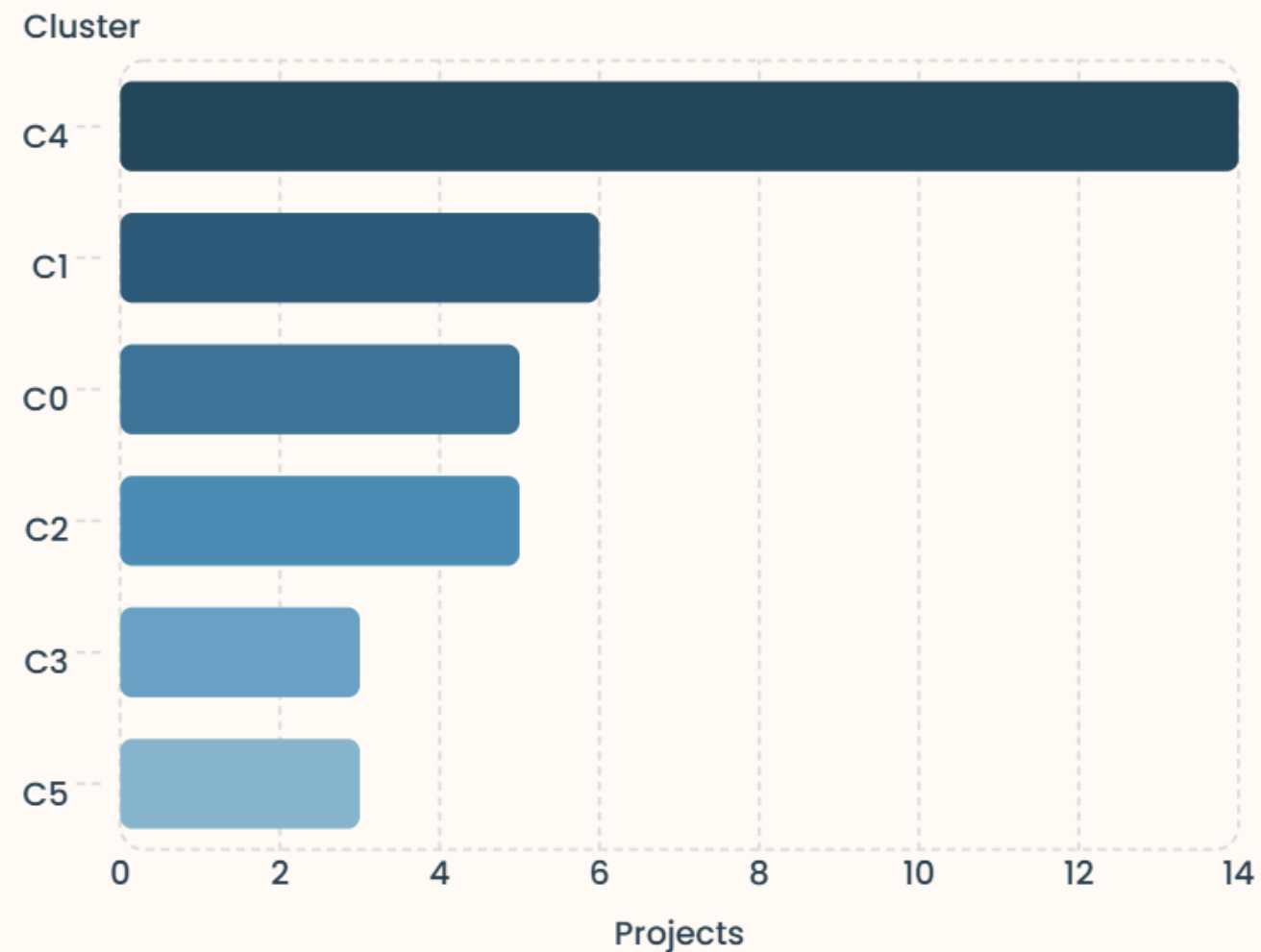
Why PCA First?

With 12 features across only 36 projects, direct clustering risks overfitting to noise. PCA compresses correlated features into orthogonal components, retaining 85.06% of variance in just 7 dimensions — enabling K-Means to identify genuinely structural groupings rather than artefacts of high dimensionality.

□ Key Result: 12 features → 7 PCs → 85.06% variance retained → 6 meaningful structural segments

Clustering Results & Validation

Six Structural Project Segments



C4 is the dominant cluster with 14 projects (38.9% of total).
C3 and C5 are the smallest with 3 projects each

Validation Metrics

0.335

Silhouette Score

Acceptable separation for a small, heterogeneous dataset

0.791

Hierarchical ARI

Strong agreement with agglomerative clustering

0.581

Bootstrap ARI

Moderate stability across resampled datasets

0.668

Composite Score

Overall validation score supporting K=6 selection

Clusters represent structural similarities, not default-risk classes. The final K=6 selection was supported through cross-method and bootstrap validation.

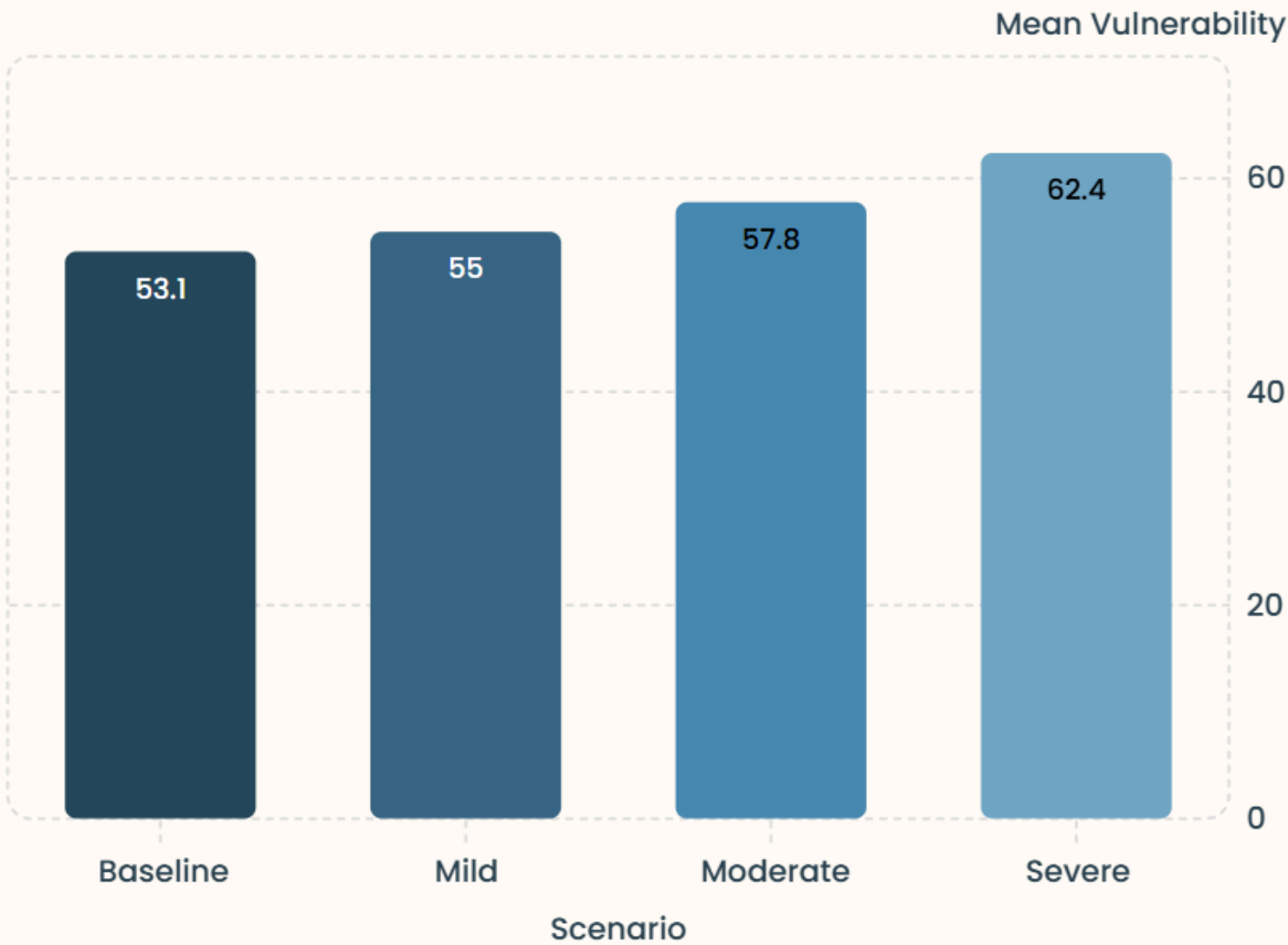
Macro Prudential Vulnerability Analysis

Vulnerability Score Components

01	Investment Scale Log-scaled capital commitment relative to sector peers
02	Technology Risk Complexity and maturity of the fabrication or packaging technology
03	Geographic Concentration State-level clustering and regional exposure
04	RBI Credit / NPA Indicators Gross NPA ratios and sectoral credit growth from RBI data
05	IIP Manufacturing Indicators Index of Industrial Production signals for macro-economic context

Only the 12 manufacturing projects enter this layer. Design\ DLI projects are excluded from stress testing due to their different risk profile

Stress Scenario Results — Portfolio Mean Vulnerability



Monte Carlo Validation

Testing the Stability of Risk Rankings

1

Stress Model

Establishes baseline scenario

2

Vary Inputs

Randomly adjust macro-financial factors

3

Run Model

Calculate portfolio vulnerability

4

Repeat 10,000x

Generate diverse outcomes

5

Distribution

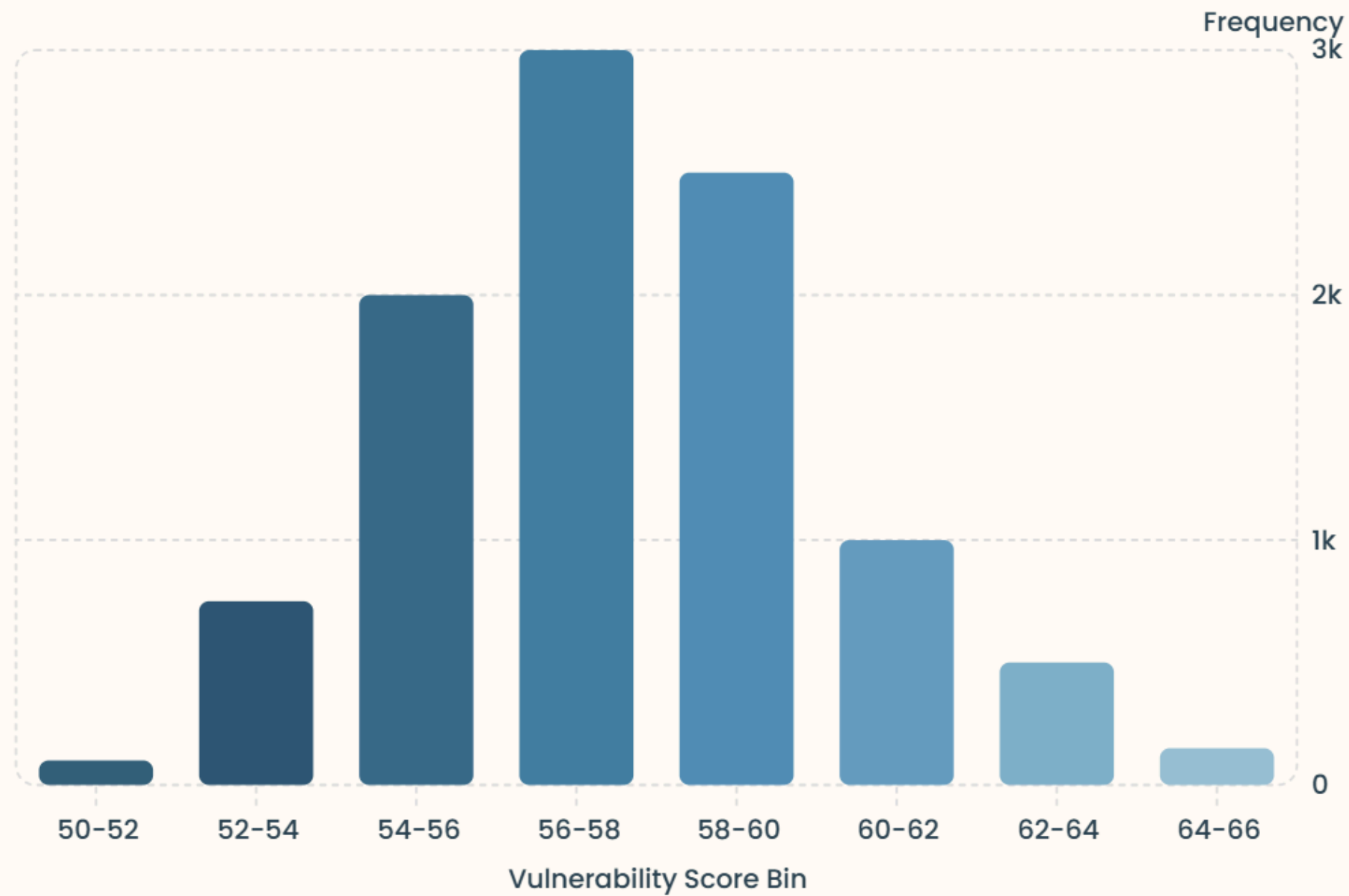
Analyze vulnerability spread

6

Measure Stability

Assess tail risk & ranking consistency

Stimulation Results



Distribution of 10,000 simulated portfolio vulnerability scores, centered around the mean with a long tail indicating potential for higher stress.

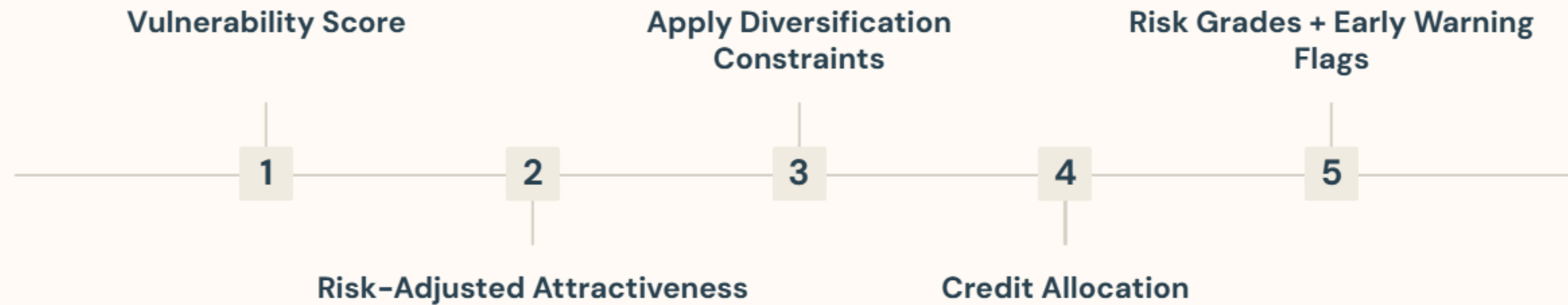
Spearman $\rho = 0.998$

This indicates extremely strong agreement between deterministic and simulated project rankings, confirming the robustness of our vulnerability assessment.

Metric	Value
Simulations	10,000
Mean	58.66
Median	58.56
P90	60.70
P95	61.30
P99	62.53
Maximum	65.75

Credit Allocation & Banking Application

From Risk Scores to Credit Decisions



Baseline Credit Allocation

Utilizing a ₹10,000 crore credit budget, the model applies dynamic constraints:

- Maximum project share: **20%**
- Maximum state concentration: **45%**
- Minimum project share: **2%**

This yields a **22.2% reduction** in portfolio stress compared to a purely proportional allocation.

Sensitivity Analysis

We tested 162 scenarios to fine-tune allocation parameters. The best configuration identified was:

- Risk weight: **80%**
- Scale weight: **20%**
- Project cap: **15%**
- State cap: **35%**

This achieved a **23.7% stress reduction** with an allocation-rank stability of Spearman $\rho = 0.892$.

Key Findings & Limitations

36

Projects Analysed

Comprehensive dataset covering India's semiconductor landscape.

6

Structural Clusters

Identified stable groupings with distinct risk profiles via K-Means.

85.06%

PCA Variance Retained

Robust dimensionality reduction preserving critical information.

10,000

Monte Carlo Simulations

Validated stress model stability and ranking consistency.

0.998

Stress-Ranking Correlation

Extremely high agreement between deterministic & simulated rankings.

0.892

Allocation-Ranking Correlation

Strong stability in risk-adjusted credit allocation.

Limitations

- **Historical Data Gap:** Insufficient historical default/NPA outcomes for empirical default modeling.
- **Not a PD Model:** Framework assesses vulnerability, not probability of default directly.
- **Rule-Based Allocation:** Current credit allocation is heuristic, not mathematically optimal.
- **Validation Pending:** Real-bank validation of the framework is still underway.

THANK YOU

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